Transformations of Functions

Name: ____________________

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Translation: a ________ of the graph — $f(x)+k$ moves it ___________, and $f(x-h)$ moves it ____________.

Reflection: a flip: $-f(x)$ flips over the _________, and $f(-x)$ flips over the _________.

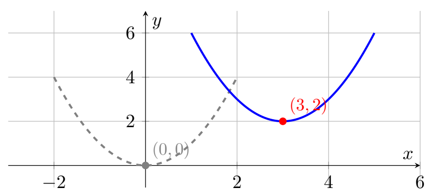
We do together. Describe how $g(x) = (x-3)^2 + 2$ transforms the parent x^2.

Step 1. Inside the square, $x - 3$ shifts the graph _________ (opposite the sign).

Step 2. The $+2$ outside shifts it ________.

Step 3. So the vertex moves from $(0, 0)$ to _________.

See it:



parent (dashed) → right 3, up 2 (solid).

You Try. Solve each. Show every step, then name the answer.

a) Describe $h(x) = -x^2$ from the parent.

b) Give the vertex of $y = (x+1)^2 - 4$.

Summary (fill in): $f(x-h)+k$ shifts a graph _______________; a leading $-f(x)$ __________ it over the x-axis.

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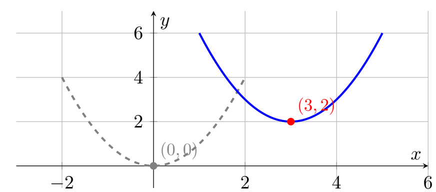

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